Jordanniel Sihombing (231112048)

SOAL 2

Jordanniel Sasr Sihombing
231112048

hal 1

3. Stegarografi

Langkah 1 : XOR utk a, x_0, c

3 huruf = Jor (ASCII 8bit)

J = 79 = 01001010

o : 111 = 01101111

r : 114 = 01110010

a = J ⊕ o : 01001010 ⊕ 01101111 = 00100101 = 37

x_0 : o ⊕ r = 01101111 ⊕ 01110010 = 00011101 : 29

c = J ⊕ r : 01001010 ⊕ 01110010 = 00111000 : 56

Langkah 2 : Rumus LCG

$$\boxed{x_{n+1} = (a \cdot x_n + c) \bmod m}$$

State utama dihasilkan dgn m = 23 utk posisi sisip, lalu nilai State ini di-mod ulang utk Channel (mod 7 → dipetakan ke R/G/B lewat mod 3) dan jumlah bit sisip (mod 4, hasil +1, agar dpt 1-4 bit)

20 State pertama ($x_1 \ldots x_{20}$), dgn x_0 = 29 ;

i	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
x	2	15	13	8	7	16	9	20	14	22	19	0	10	12	17	18	9	21	5	11

- Channel (x mod 7 → mod 3 utk RGB = 2, 1, 6, 0, 2, 4, 6, 0, 1, 5, 0, 3, 5, 3, 4, 2, 0, 5, 4,

mod 3 → B, G, B, G, R, B, G, R, B, G, R (?)

- Jumlah bit Sisip (x mod 4 +1) : 3, 4, 2, 1, 4, 1, 1, 1, 3, 3, 4, 1, 3, 1, 2, 3, 2, 2, 4.

Langkah 3 : Pesan

Left (Jordanniel, 5) huruf kapital : "JORDA"

J = 01001010 O : 01001111 R : 01010010 D : 01000100

A : 01000001

bit stream (40 bit) : 0100101001001111010100100100010001000001

Forte

Langkah 4 : Cover Image 5x5 (Nilai bukan unik)

Pixel	R	G	B
P0	50	80	120
P5	60	87	133
P10	70	99	146
P15	80	101	159
P20	90	108	172

Langkah 5 : Tabel Proses Penyisipan (LSB Replacement)

~~Pixel Channel Jlh bit Bit sisip Nilai baru~~

	Pixel	Channel	Jlh bit	Bit sisip	Nilai lama	Biner lama	Nilai baru	Biner baru
1	P2	B	3	010	130	10000010	130	10000010
2	P15	G	4	0101	101	01100101	101	01100101
3	P13	R	2	00	79	01001111	76	01001100
4	P8	G	1	1	120	01111000	121	01111001
5	P7	R	4	0011	66	01000010	67	01000011
6	P16	B	1	1	164	10100100	165	10100101
7	P9	G	1	1	124	0111100	125	01111101
8	P20	R	1	0	90	01011010	90	01011010
9	P14	R	3	101	82	01010010	85	01010101
10	P22	G	3	001	130	10000010	129	10000001
11	P14	B	4	0010	179	10110011	178	10110010
12	P0	R	1	0	50	00110010	50	00110010
13	P10	R	3	010	70	01000110	66	01000010
14	P12	B	1	0	156	10011100	156	10011100
15	P17	R	2	01	86	01010110	85	01010101
16	P13	G	3	000	134	10000110	128	10000000
17	P9	B	2	00	153	10011001	152	10011000
18	P4	R	1	1	93	01011101	93	01011101

Total 40bit Pesan tersisip habis pd Langkah ke-18.

Langkah 6 : Load capacity

pixel ~~stl~~ untuk sisipan : 18 dari 25 pixel total

~~Load~~ Load capacity $= \frac{18}{25} \times 100\% = 72\%$

Langkah 7 : PSNR

$MSE = \frac{1}{N} \sum (\text{cover} - \text{stego})^2 = \frac{78}{75} = 1.04$

N = 25 pixel × 3 channel = 75 nilai :
total selisih kuadrat = 78

$PSNR = 10 \log_{10}\left(\frac{255^2}{MSE}\right) = 10 \log_{10}\left(\frac{65025}{1.04}\right) = \underline{47.96 \text{ dB}}$

PSNR > 40 dB , menunjukkan kualitas citra stego sangat baik.